Excel-Task

This repository contains my Microsoft Excel tasks and Practical projects focused on data analysis and visualization. It showcases my ability to clean, organize, analyze, and interpret data using Excel.

📊 Data Exploration Using Excel

📌 Project Overview

This project focuses on exploring and analyzing a product dataset using Microsoft Excel. It demonstrates how Excel functions can be used to summarize product prices, identify minimum and maximum values, classify products based on price, calculate conditional totals, and extract information from Product IDs.

The project uses a product dataset containing 34 records with details such as Product ID, Product Name, Brand Name, Price, Quantity, and Category.

🎯 Project Objectives

Calculate total, count, and average product prices.

Identify minimum and maximum product prices.

Classify products using the IF function.

Calculate category-wise totals using SUMIF.

Count products meeting specific price conditions using COUNTIF.

Extract Day, Country Code, and Month from Product IDs.

Develop practical Excel skills for data exploration and analysis.

🛠️ Tools Used

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🛠️ Tools Used

Microsoft Excel

Excel Formulas and Functions

Product Dataset

📁 Dataset Description

The dataset contains the following columns:

Column	Description
Product ID	Unique or identifying code associated with a product record
Product Name	Name of the product
Brand Name	Brand or manufacturer
Price ($)	Price of the product in dollars
Quantity	Available or recorded product quantity
Category	Product category

Dataset size: 34 records and 6 original columns.

Note: The dataset contains repeated records. All calculations in this project include every row provided, including duplicates.

1. ➕ SUM, COUNT, and AVERAGE

Perform basic calculations using Excel functions.

A. Calculate the Total Price of All Products

Use the SUM function to calculate the total of all product prices.

Excel Formula:

=SUM(D2:D35)

Expected Result: $10,100

Explanation: Adds all numeric values in the Price column.

B. Determine the Number of Products

Use COUNTA to count the product records using Product ID.

Excel Formula:

=COUNTA(A2:A35)

Expected Result: 34 records

Explanation: Counts all non-empty Product ID cells.

Alternatively, use the COUNT function on the numeric Price column:

=COUNT(D2:D35)

Expected result: 34

C. Calculate the Average Price

Use the AVERAGE function to calculate the average price of the products.

Excel Formula:

=AVERAGE(D2:D35)

Expected Result: $297.06

Explanation: Calculates the sum of the prices divided by the number of numeric price entries.

2. 📉 Minimum and Maximum Values

Use Excel functions to identify the lowest and highest product prices.

A. Identify the Minimum Price

Excel Formula:

=MIN(D2:D35)

Expected Result: $30

Product: T-shirt — Adidas

B. Identify the Maximum Price

Excel Formula:

=MAX(D2:D35)

Expected Result: $1,000

Product: Laptop — Dell

Purpose: These functions help identify the least expensive and most expensive products in the dataset.

3. 🧠 Logical Function – IF

Create a new column named Price Range to classify products according to their price.

Business Rules

If Price is greater than or equal to $500, classify it as High Price.

If Price is less than $500, classify it as Standard Price.

Excel Formula

Add Price Range as a new header in cell G1.

Enter the following formula in cell G2:

=IF(D2>=500,"High Price","Standard Price")

Drag the formula down to cell G35.

Example Output

Product Name	Price ($)	Price Range
Laptop	1000	High Price
Sneakers	80	Standard Price
Coffee Maker	130	Standard Price
Smartphone	900	High Price
Tablet	500	High Price
T-shirt	30	Standard Price

Purpose: The IF function helps categorize products based on business rules and makes the dataset easier to analyze.

4. 🧮 Conditional Functions – SUMIF and COUNTIF

A. SUMIF – Total Price of Electronics Products

Calculate the total price of all products belonging to the Electronics category.

Excel Formula:

=SUMIF(F2:F35,"Electronics",D2:D35)

Expected Result: $8,050

Explanation:

F2:F35 — Category range.

"Electronics" — Condition to match.

D2:D35 — Price values to sum.

Purpose: This function helps analyze the total price of products within a specific category.

B. COUNTIF – Count Products Priced Below $100

Determine how many product records have a price below $100.

Excel Formula:

=COUNTIF(D2:D35,"<100")

Expected Result: 11 records

Explanation: Counts every record whose Price value is less than $100.

Purpose: This function helps identify and count lower-priced products.

5. 🔤 Text Formatting – LEFT, RIGHT, and MID

Use Excel text functions to extract information from the Product ID column.

The Product ID format in this dataset is:

DD-MMM-CC

For example: 28-JAN-US

DD represents the day code.

MMM represents the month code.

CC represents the country code.

A. LEFT – Create the Day Column

Create a new column named Day in cell H1.

Extract the first two characters from each Product ID.

Excel Formula:

=LEFT(A2,2)

Example:

Product ID	Day
28-JAN-US	28
15-FEB-US	15
03-MAR-US	03
11-APR-US	11

Explanation: The LEFT function extracts the first two characters from the Product ID.

B. RIGHT – Create the Country Code Column

Create a new column named Country Code in cell I1.

Extract the last two characters from each Product ID.

Excel Formula:

=RIGHT(A2,2)

Example:

Product ID Country Code

28-JAN-US US

15-FEB-US US

07-JUN-UK UK

17-JUN-IN IN

08-DEC-DE DE

Explanation: The RIGHT function extracts the final two characters, representing the country code in this dataset.

C. MID – Create the Month Column

Create a new column named Month in cell J1.

Extract the 4th to 6th characters from each Product ID.

Excel Formula:

=MID(A2,4,3)

Example:

Product ID	Month
28-JAN-US	JAN
15-FEB-US	FEB
03-MAR-US	MAR
11-APR-US	APR
22-MAY-US	MAY
07-JUN-UK	JUN

Explanation: The MID function starts at character position 4 and extracts 3 characters, returning the month abbreviation.

Final Dataset Structure

After adding the new columns, the dataset will contain the following fields:

Column	Field Name
A	Product ID

B Product Name

C Brand Name

D Price ($)

E Quantity

F Category

G Price Range

H Day

I Country Code

J Month

Copy the formulas down through row 35 to apply them to all 34 records.

6. 📋 Summary of Excel Functions

Function Purpose Formula

SUM Total product prices =SUM(D2:D35)

COUNT Count numeric price values =COUNT(D2:D35)

COUNTA Count non-empty Product IDs =COUNTA(A2:A35)

AVERAGE Average product price =AVERAGE(D2:D35)

MIN Minimum product price =MIN(D2:D35)

MAX Maximum product price =MAX(D2:D35)

IF Classify products by price =IF(D2>=500,"High Price","Standard Price")

SUMIF Total price of Electronics products =SUMIF(F2:F35,"Electronics",D2:D35)

COUNTIF Count products priced below $100
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LEFT Extract the first two characters =LEFT(A2,2)

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MID Extract the month abbreviation =MID(A2,4,3)

7. 📊 Key Findings

Based on the supplied dataset, the following results were obtained:

Metric	Result
Total price	$10,100
Number of product records	34
Average price	$297.06
Minimum price	$30
Maximum price	$1,000
Electronics total price	$8,050
Products priced below $100	11

Note: Total price and average price are calculated from the Price column only, not by multiplying Price by Quantity. Duplicate records are included in all calculations.

8. 🎓 Key Learnings

Through this project, I practiced:

Basic statistical analysis using SUM, COUNT, AVERAGE, MIN, and MAX.

Logical decision-making using IF.

Conditional calculations using SUMIF and COUNTIF.

Text extraction using LEFT, RIGHT, and MID.

Adding calculated columns to an existing dataset.

Applying Excel formulas to a practical product dataset.

Summarizing data to identify useful insights.

9. 🚀 Future Improvements

Analyze total inventory value using Price × Quantity.

Compare product prices across categories and brands.

Explore sorting, filtering, and conditional formatting.

Create PivotTables and PivotCharts.

Build an interactive Excel dashboard.

Extend the analysis using SQL, Python, and Power BI.

👨‍💻 Author

A Lakshmana Narayanan Aspiring Data Analyst

This project is part of my Data Analytics learning journey, focusing on Excel functions, data exploration, and practical analysis.

⭐ Acknowledgment

Created for educational purposes to develop practical Microsoft Excel skills and establish a foundation for more advanced data analytics projects.Quantity Available or recorded product quantity

Category Product category

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